

Appendix A

Describing the targets workflow and presenting some supplementary results for the paper titled ‘Bias Correction and Uncertainty Quantification for Small-Sample Ecosystem Condition Assessments’

The analyses are arranged into a pipeline, or targets workflow. The workflow file is called `__targets.R`, and it relies on functions described in the file `R/functions.R`. All the code is available on GitHub at <https://github.com/anders-kolstad/HIAs> with an archived stable version on Zenodo (doi: [10.5281/zenodo.1000000](https://doi.org/10.5281/zenodo.1000000)). When running the workflow, the individual targets (outputs from functions) gets stored in a folder locally. The manuscript, as well as this appendix document, is written in quarto. We can then read the targets back into R and print or visualise them, which is what we have done in this current document, and we leave the R code visible to be explicit about where in the targets workflow we are extracting information.

Figure [1](#) shows the target named `test_most_common`, which is a plot showing the most common wetland nature types in Norway.

```
tar_read(test_most_common)
```

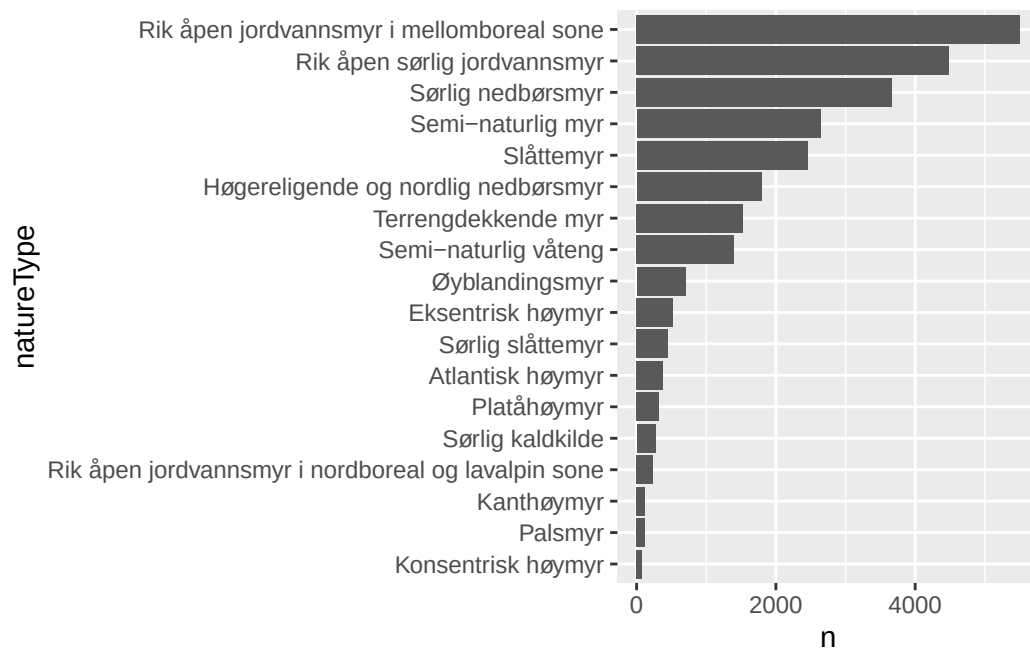


Figure 1: Ordered barplot showing the most common wetland nature types in Norway in terms of occurrence frequency.

Figure 2 shows the distribution of variable values for the five variables extracted from the nature types dataset, after they have been converted to a percentage scale.

```
tar_read(test_naturetypes_p)
```

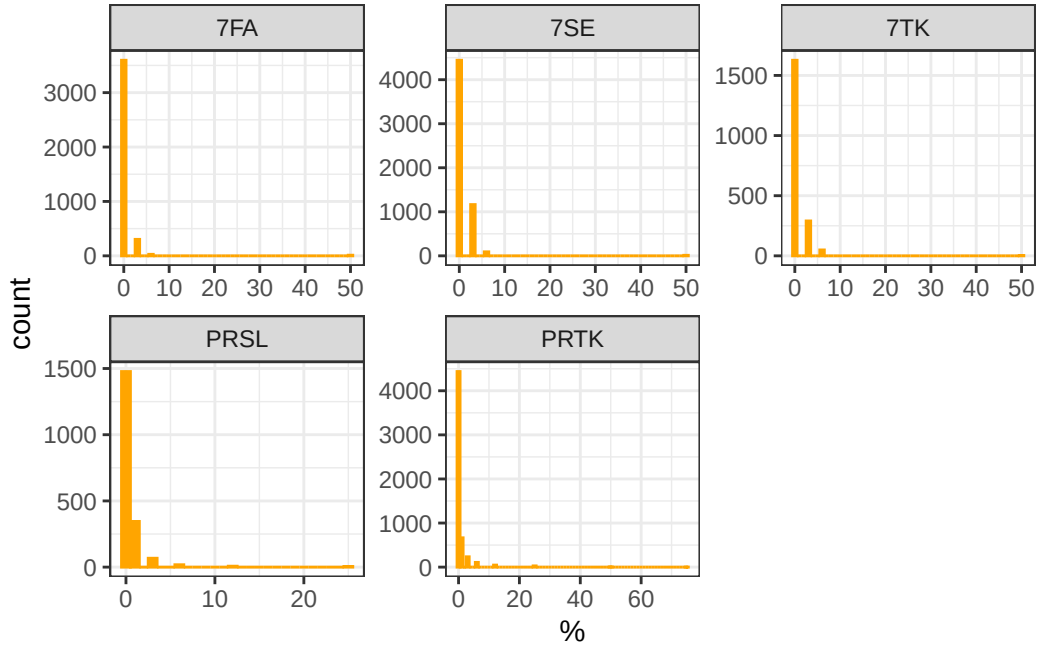


Figure 2: Distribution of variable values.

The following snippet from the target *naturetypes_comb* shows how nature type polygons (rows) typically don't have all four of the variables underlying the ADSV indicator recorded. *7SE* and *PRSL* are combined into the variable *SE*, and *PRTK* and *7TH* are combined into the variable *TK*. Then the sum of *TK* and *SE* makes up the variable *ADSV*.

```
tar_read("naturetypes_comb") |> head()
```

```
# A tibble: 6 x 8
```

```
# Rowwise:
```

	id	var_7SE	var_PRTK	var_7TK	var_PRSL	TK	SE	ADSV
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	NINFP2310142406	0	0	NA	NA	0	0	0
2	NINFP2010014477	3.12	0	NA	NA	0	3.12	3.12
3	NINFP1910019329	0	1.5	NA	NA	1.5	0	1.5
4	NINFP2010021023	NA	NA	0	1.5	0	1.5	1.5
5	NINFP2010016336	0	0	NA	NA	0	0	0
6	NINFP1910006253	0	0	NA	NA	0	0	0

To combine these four variables into one metric, ADSV, one could take the one with the highest value (worst-rule), or the sum. Summation is problematic as not all locations have two values

to sum together. But the other option is problematic since field workers often tend to split the effects over two variables, if they have that option, so that the summed variable values matches what they perceive as the correct answer. If we have 50% vehicle damage and 50% hiking damage, that is no doubt worse than having 50% of just one of them. So I will use the sum, despite its issues.

Figure 3 shows the distribution of the two pairs of related variables after taking their sum, and Figure 4 show the distribution after summing the two variables in Figure 3.

```
tar_read(plot_naturetypes_comb)
```

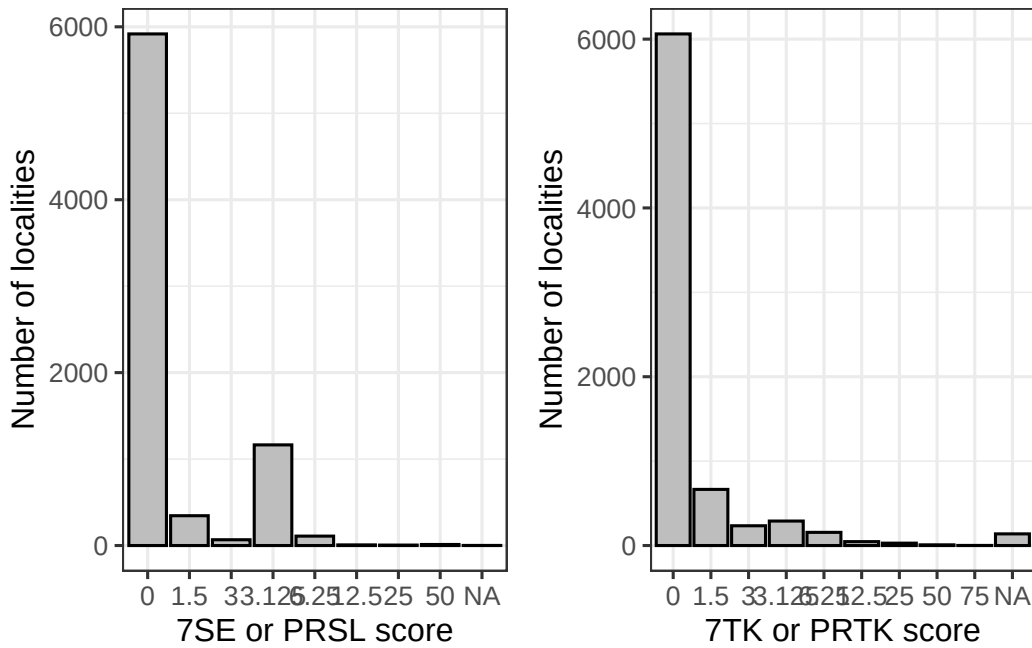


Figure 3: Distribution of summed variable values for the two pairs of related variables 7SE/PRSL and 7TK/PRTK.

```
tar_read(plot_naturetypes_comb)
```

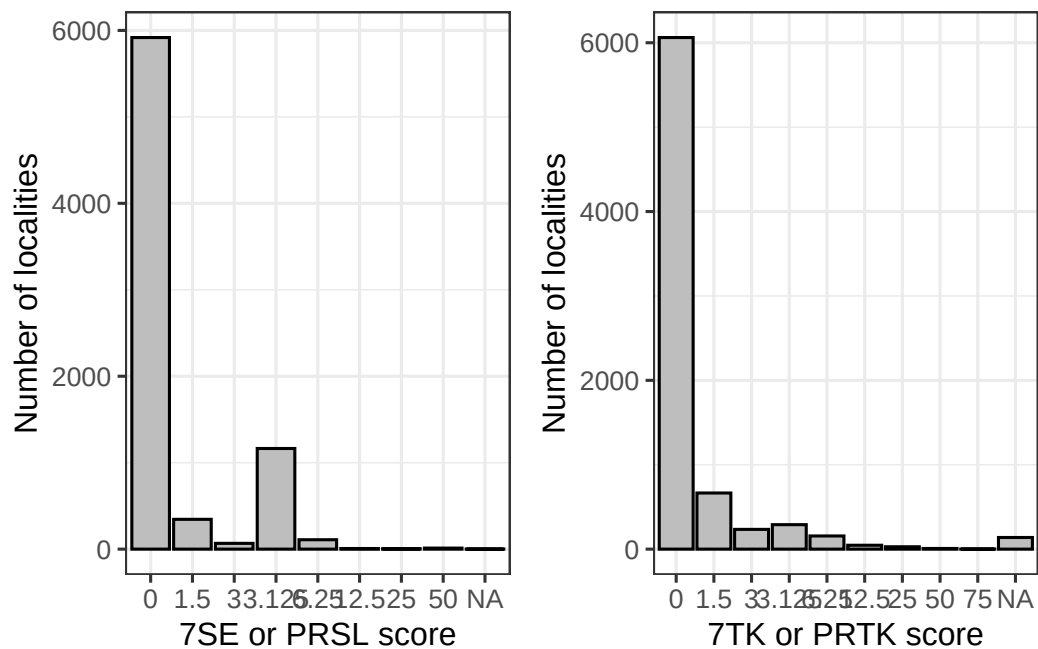


Figure 4: Distribution of summed ADVS variable values.

Figure 5 shows the normalisation of three variables into a dimensionless indicator scale. The scaling functions are non-linear for ADSV and 7FA, ensuring that 0.6 on the indicator scale corresponds to the variable values that marks the threshold between good and reduced ecosystem condition.

```
tar_read(plot_normalised)
```

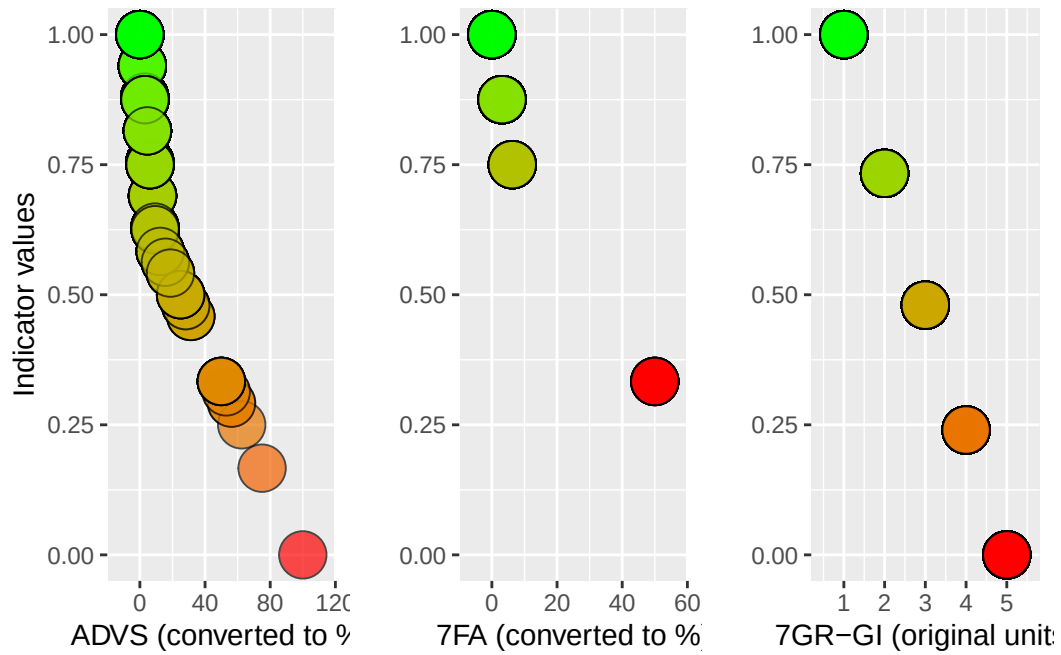


Figure 5: Scaling function for normalising the three condition variables.

Figure 6 show the location of the three example municipalities included in this study.

```
positionMap <- tar_read(positionMap_png)
knitr::include_graphics(positionMap)
```

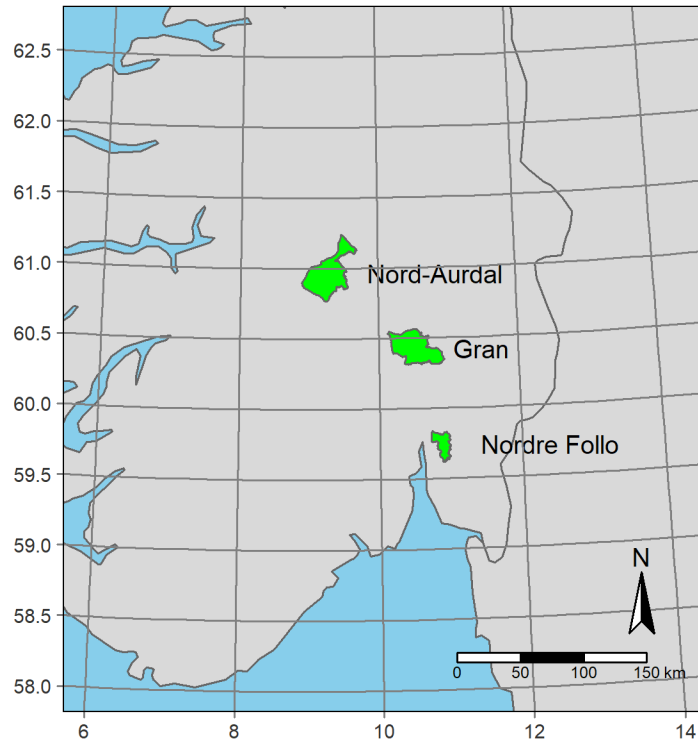


Figure 6: Position of the three municipalities analysed in this study.

Figure 7 show how the indicator values are dispersed in Nord Aurdal municipality before they are extrapolated across to other mire occurrences in the municipality as in Figure 5.

```
indicator_magnify_png <- tar_read(indicator_magnify_png)
knitr::include_graphics(indicator_magnify_png)
```

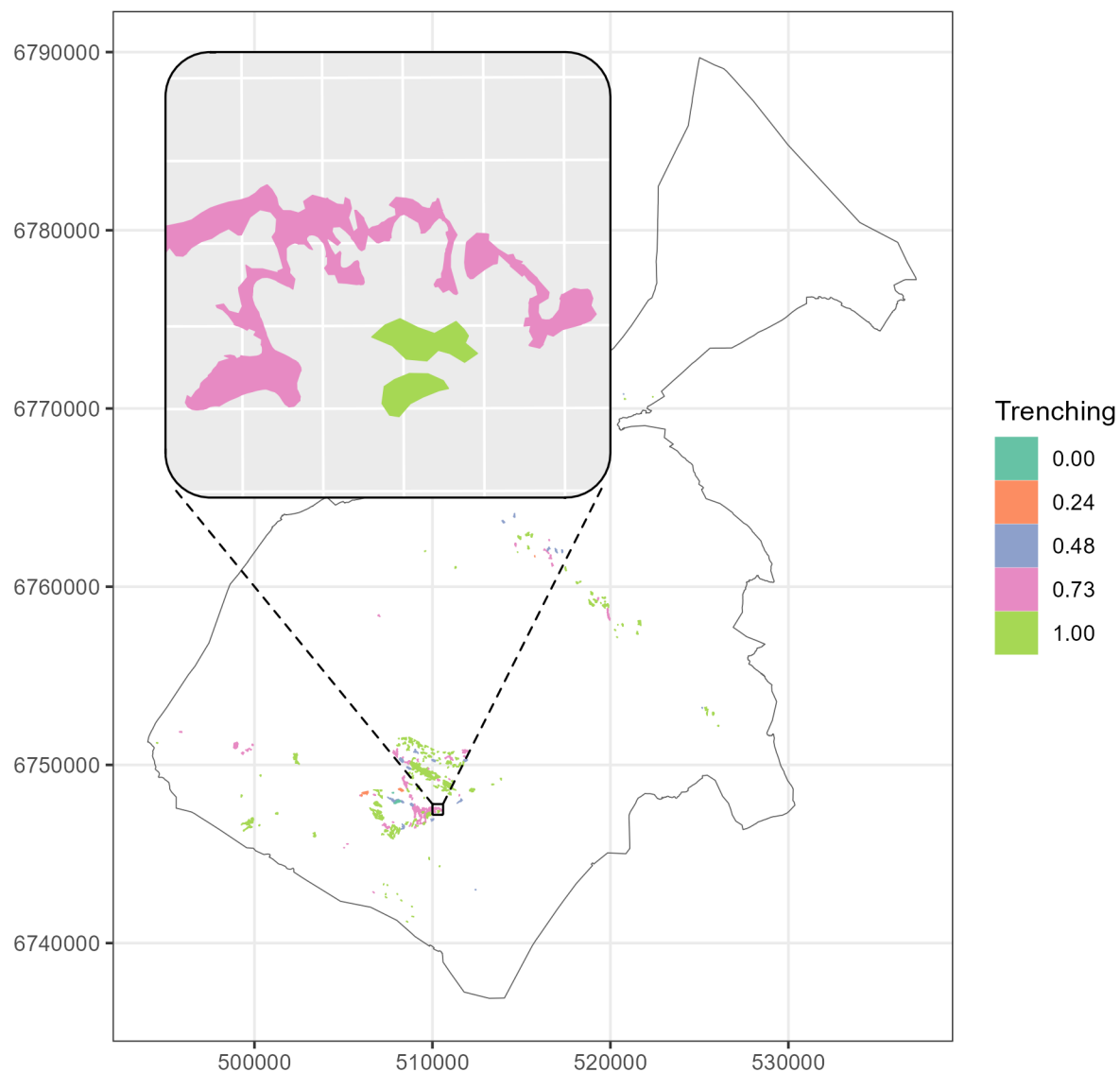


Figure 7: Map of Nord Aurdal municipality with surveyed mires colored to show the indicator values.

Table 1 shows the same information about the indicator values as Figure 6, but in a table.

```
knitr::include_graphics(here::here("output", "fig", "EEA_tbl_screenshot.png"))
```


Table 1: Indicator values and 95% credible intervals for three indicators in three municipalities.

Indicator	Indicator value
Gran	
ADSV	0.74 [0.66 - 1]
Alien species	0.9 [0.77 - 1]
Trenching	0.42 [0.35 - 1]
Nord-Aurdal	
ADSV	0.9 [0.71 - 1]
Alien species	0.95 [0.75 - 1]
Trenching	0.89 [0.43 - 1]
Nordre Follo	
ADSV	0.98 [0.71 - 1]
Alien species	0.9 [0.79 - 1]
Trenching	0.25 [0 - 1]